



Object Oriented Design and Modeling through UML Past Questions

Object Oriented Design and Modeling through UML (Pokhara University)



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POKHARA UNIVERSITY

Level: Bachelor

Semester : Spring

Year : 2023

Programme: BE

Full Marks: 100

Course: Object Oriented Design and Modeling
through UML

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.
 - a) Explain the difference between Object-Oriented Analysis and Object-Oriented Design with an example. 7
 - b) Why is use case description important? Provide an example and discuss about the important section of a use case description. 8
2.
 - a) Discuss about the different phases of unified process and their role during software development. 8
 - b) What is a contract? Explain different parts of contract with suitable example. 7
3.
 - a) What is the purpose of developing a conceptual model? How is a domain model represented? Discuss its components. 7
 - b) "Designers make a bridge between customer requirements and the actual working code", Justify this statement using major sets of design artifacts. 8

4. Case Study

Online Medical Store system allows us access their system with a valid mobile number. We can search for medicines in their systems by providing the doctor's prescription list. The prescription list is compared and a list of available medicines is displayed to us. Then we select the medicines and proceed to checkout and make payment. We also indicate the address for delivery. The system displays the expected delivery date. After that, a SMS with the delivery details is sent to our mobile number. Also, the system facilitates the customers to cancel their order 1 week before delivery with no cancellation charge. After 1 week no cancellation can be made.

Draw a use case model, Domain model, collaboration diagrams and design class diagram from these information.

15

5. a) Discuss the scope and visibility of objects. Elaborate the use of parameter visibility and local visibility in details. 7
b) How cohesion and coupling affect a good design? Explain with suitable points. 8
6. a) Explain about Fault, error and failure with relevant examples. Discuss the steps to handle exception in object oriented programming. 8
b) Designs are for programmers not for end users. Show how different design diagrams help programmers for writing code. 7
7. Write short notes on: (Any two) 2×5
a) Conceptual class category list
b) Collection Classes
c) Information expert

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year : 2021

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through UML

Pass Marks: 45

Time : 3hrs.

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Attempt all the questions.

1. ☒ a) How does OOA differ from OOD? Elaborate your answer with key activities in each phase. 7
- ☒ b) What are use cases and how they can be used for identifying and recording system requirements? How would you differentiate between include and extend relationship between use cases? 8
2. ☒ a) What are different strategies for identifying conceptual classes? Explain with examples. 8
- ☒ b) What is an operation contract? Is it essential to develop an operation contract for each and every system operations? Explain. 7
3. ☒ a) What is visibility? Describe the use of attribute and parameter visibility among classes while designing class diagrams. 8
- ☒ b) Discuss the necessity of interaction diagram during software design. How is conditional messages and mutually exclusive conditional paths illustrated in interaction diagrams? Explain. 7
4. ☒ a) Explain different types of relationship used in class diagram. 8
- ☒ b) "UML is a language for specializing, visualizing, constructing and documenting the artifacts of the system." Justify this statement explaining the significance of UML in OOA/D. 7
5. ☒ a) What is the significance of studying exception and error handling in objects oriented analysis and design? Explain briefly about exception handling technique. 7
- ☒ b) Explain the process of mapping design to code (both class definition and method definition) with example. Support your answer by using a design class diagram and collaboration diagram. 8

6. Online Learning platforms help students to search for courses as per their criteria. If the system shows a suitable course then the students registers for the course and paying for it. Interaction with teacher can be done by registered students. Students need to attend the class and submit their assignments regularly. Students who attended 80% of the classes and submitted all their assignment are awarded with a course completion certificate at the end of the course. Also, the system facilitates the students to cancel their registration and refund the payment if it is done within a week. Now draw a use case model (no description required), Domain model, interaction diagrams and design class diagram from these information.

15

7. Write short notes on: (Any two)

-2×5

- a) Functional Vs. Non-Functional Requirements
- b) Coupling and cohesion
- c) Reverse Engineering

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year : 2020

Programme: BE

Full Marks: 100

Course: Object Oriented Design and Modeling
through UML

Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. ☒ (a) How does OOD differ from OOA? Explain the key OOA activities with example. 7
- ☐ (b) Explain different phases and disciplines of Rational Unified Process. 8
2. ☒ (a) What do you mean by UML? Draw a use-case diagram for a library management system and write fully dressed use case for "issue-book". 8
- ☐ (b) Explain the importance of object-oriented analysis, in identifying conceptual classes and other elements and to construct a domain model, with a scenario. 7
3. ☐ (a) What is the role of Domain Model in developing a software? How can you construct a domain model? 7
- ☐ (b) How does classes are linked to one another? Discuss association with its name and multiplicity with examples. 8
4. **Case Study: Course Registration System**

At the beginning of each semester students may request a course catalog containing a list of each course offerings for the semester. Information about each course, such as professor, department, and prerequisites will be included to help students for prompt decisions. The system will allow students to select four courses that are going to be offered for the coming semester. In addition, each student will indicate two alternative choices in case a course offering becomes filled or canceled. No course offerings will have more than ten students or fewer than three students. A course offering with fewer than three students will be canceled. Once the registration process is completed for a student, the registration system sends information to the billing system so the student can be billed for the semester. Professors must be able to access the online system to indicate which courses they will be teaching, and to see which

students signed up for their course offerings.

- a) Considering above scenario, draw the collaboration diagram. 7
- b) Identify the classes, assume the necessary attributes and draw the class diagram. 8
5. a) What is visibility in OOP Language? Describe its role in obtaining high abstraction. What are the various types of them? Elaborate using diagram and code snippets for the private visibility. 8
- b) Why do we need design patterns? Explain the use of low coupling and high cohesion patterns. 7
6. a) How can we make our system more robust? Discuss the use of exception handling in OOA and OOD. 7
- b) Ultimately diagrams are not executable, so we need to convert them into code. Explain necessary steps to convert design to code with example. 8
7. Write short notes on: (Any two) 2×5
 - a) Elaborating use case
 - b) Representation of system behavior
 - c) Mapping design to code

POKHARA UNIVERSITY

Level: Bachelor
Semester: Fall
Programme: BE
Course: Object Oriented Design and Modeling
through UML

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

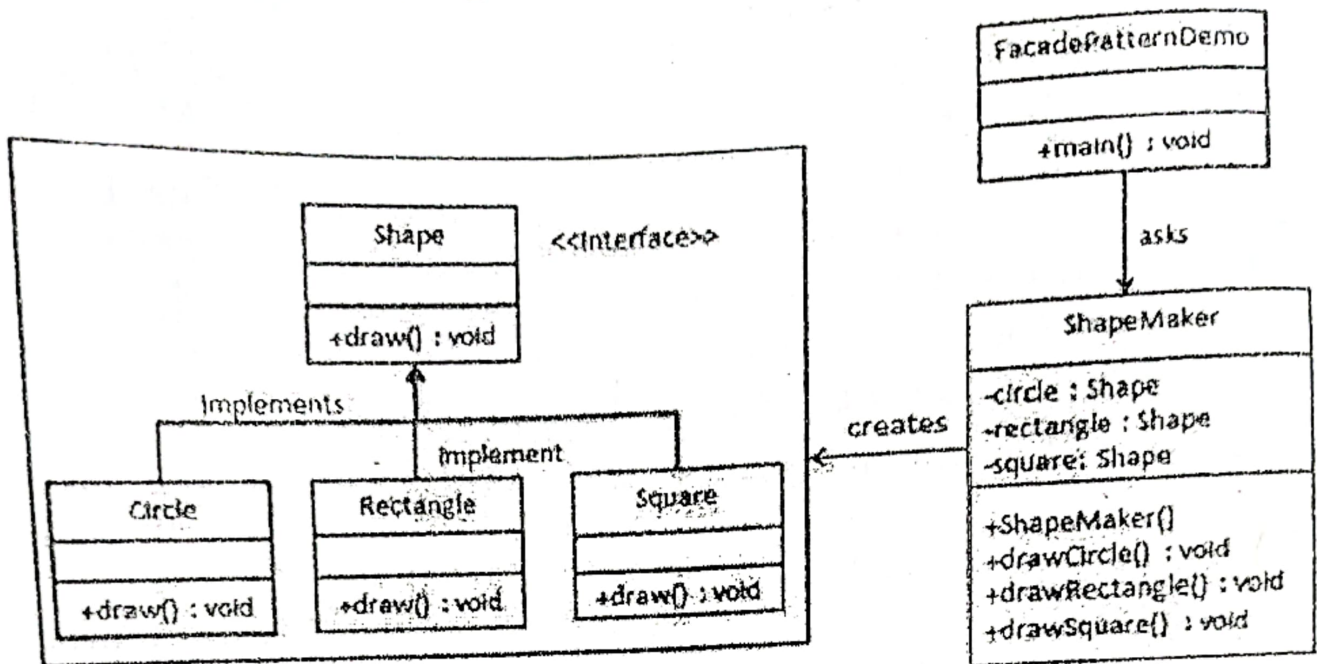
Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. (a) "Unified Modeling Language (UML) is a language for specifying, visualizing, constructing, and documenting the artifacts of software systems". Justify this statement explaining the significance of UML in OOA/D. 8
- (b) What are the various formats to represent a Use case? Explain Fully-Dressed format with an example. 2+5
2. a) How System Behaviour can be represented in UML? 8
- b) What is conceptual Model? How can one build a conceptual model? Explain the steps involved. 7
3. a) Describe the significance of GRASP. Explain Information Expert Pattern with an example. 7
- b) How do you denote mutually exclusive conditional paths in Interaction Diagram? What are the DO-ing and KNOW-ing responsibilities of an object? 8
4. a) What is a collaboration diagram? How do we represent mutually exclusive conditions in a collaboration diagram? Justify with a complete block example. 7
- b) What do you mean by system operation, events and behaviour? Draw a SSD for makePayment() function. 8

- a) Convert the given Design Class Diagram into an equivalent code. (you can choose any language like Java, C++, etc)



- b) What do you mean by Exception in OOA/D? How are Exceptions addressed by UML?

6. a) How object oriented is used in SDLC?
 b) "Contracts in UML provides a more detailed description of system behavior when use cases are insufficient". Explain this statement with an example.

7. Write short notes on: (Any two)

- a) System event and system operation illustrating un UML
 b) Visibility
 c) High Cohesion and Low Coupling

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
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Through UML

Semester: Fall

Year : 2018
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. (a) Explain data and behavioral abstraction with suitable examples? 8
How is the responsibilities assigned to objects different from the behavior that they exhibit?
- (b) Explain the process of performing requirements elicitation and specifying those requirements using a visual modeling tool? 7
2. (a) What are use cases and how can they be used for identifying and recording system requirements? How would you differentiate between include and extend relationship between use cases? 8
- b) What are the different relationships that may exist between Domain concept elements? Construct a domain model for a patient care system making necessary assumptions. 7
3. a) How are system events identified? 9
Student Attendance Monitoring System:
In case of a Student Attendance Monitoring System, the system allows students to record their daily attendance through a Biometric Finger scan device. The system records the daily attendance of students and generates a report at the end of every month that shows the late arrival details, number of absentees etc. The system allows staff members to print the reports and configure the system as per their need.
Identify the potential use cases for scenarios and elaborate them further by coming up with individual sequence diagrams for identifying the system events belonging to the individual scenario.
- b) How can iterations and conditional messages be shown using a Collaboration diagram and a Sequence diagram? 6
4. a) **Cheque Payment:** 8
A customer presents a cheque to the bank employee. The employee checks if the cheque and the signature are valid and that the

customer has enough amount of balance in the account. The maximum amount of withdrawal allowed is 100000 per cheque unless the customer is a Privileged Customer in which case the customer can make a withdrawal of 500000 per cheque. Upon successful verification, the system records the successful completion of the transaction and the cash is handed over to the customer.

- i. Perform domain modeling of the above scenario.
 - ii. Draw the sequence diagram for the events that occur in the scenario.
- b) What is the difference between coupling and cohesion? How can you make use of the Information Expert Pattern rather than the Controller pattern for delegating responsibilities to objects while designing a system? 7
5. a) In the UML, how can error generated and exceptions thrown by operations be illustrated in a Sequence diagram? Explain with the necessary figure. 7
- b) **ATM:** 8
A customer having a Savings account in a Bank swipes a card in an ATM machine and types in the PIN number using the keyboard of the ATM. The System checks if the PIN matches the card. If the PIN does not match, the system rejects the card and display the rejection message in the ATM display. If the PIN is found to be a valid one, the customer types in the amount to be withdrawn. The system checks if the customer has enough amount of balance in that account. If he does, the system debits the account and dispatches cash thru the dispenser of the ATM and a receipt gets printed and issued to the customer.
Come up with a design class diagram for the scenario above and then map those classes to code in any programming language that supports Object Oriented Paradigm.
6. a) What are the artefacts created during the Object Oriented Analysis phase of software development and how do those artefacts get converted into concrete design during the Object Oriented Design phase thru elaboration? 8
- b) Why should one expect and plan for change and deviation from the design even during programming? Explain. 7
7. Write short notes on: (Any two) 2
- a) Activity Diagrams
- b) Iterative Process
- c) Local and Global Visibility

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Through UML

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Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. (a) Why is it necessary to form abstractions? Explain the process of forming data and behavioural abstraction with suitable examples? 8
(b) Explain the difference between requirements specification and requirements elicitation? How are they performed? 7
2. (a) Differentiate static and dynamic model used in software development life cycle with a concise diagram of each. 7
b) **Yummy Online Food Ordering:** 8
In case of an Online Food Ordering System, the system allows customer to order food online. The system records the orders placed by the customers and generates a report at the end of every month that shows the order details. The system notes the customers who place orders more frequently and provides discount as per the published discount scheme. It also allows the system administrator to configure the system and print reports.
For the above mentioned scenario, come up with the potential use cases and elaborate them further by coming up with individual sequence diagrams for identifying the system events belonging to the individual scenario.
3. a) How do you come up with the Domain concept elements from the requirements specified by the customer? Your college is wanting to automate the manual tasks being conducted in the Exam section, come up with a domain model for this scenario by making the necessary assumptions. 7
b) Explain system response, event and operation. Explain the purpose of SSD with an example. 8
4. a) How do the artifacts created during the Object Oriented Analysis 7

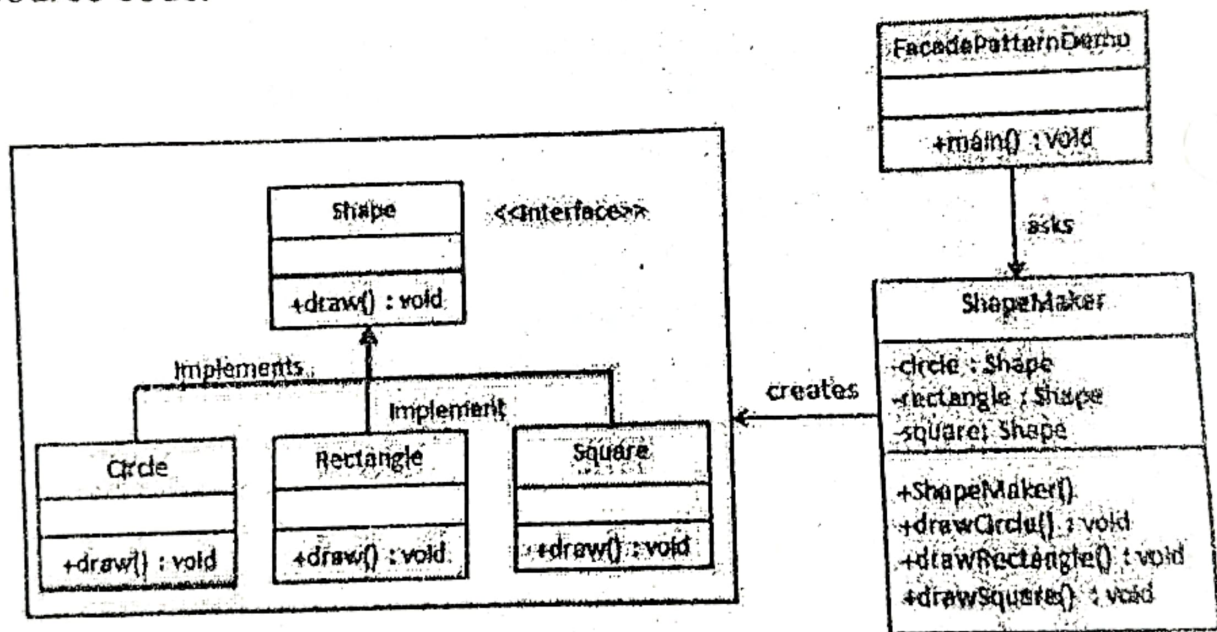
phase of software development get converted into concrete design during the Object Oriented Design phase? Explain with an example.

b) Differentiate between coupling and cohesion by drawing class diagrams of interconnected classes? When would you rather make use of the Information Expert Pattern than the Creator pattern for responsibility delegation to objects while designing an Object Oriented system?

a) Differentiate between sequence and communication diagrams with regards to their strength and weakness. Draw a diagram for mutually exclusive conditional path.

b) What is class diagram? How class diagram can be drawn? Explain step by step methodology along with the notations used to create them.

6. a) On the basis of given figure, convert it to code using a complete source code.



b) Differentiate between errors and exceptions. What features are provided by the UML for modeling exception handling?

7. Write short notes on: (Any two)

a) Steps to create a domain model.

b) Visibility of objects

c) Creating class definitions from DCD's.

POKHARA UNIVERSITY

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Time : 3hrs.

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Attempt all the questions.

1.

a) Explain the difference between Object Oriented Analysis and Object Oriented Design with an example? 7

b) Explain the role of UML model in software design? Explain types of use cases with examples. 8
2.

a) Explain difference between incremental and iterative software development model. 7

b) Explain system event and system response? Explain the purpose of SSD with an example. 8
3.

a) Explain the guidelines for preparing a contract with its purpose. 7

b) What is Global visibility how it is different from parameter visibility, explain with example. 8
4.

a) Define conceptual class. Explain the role of domain modeling with an example. 8

b) Explain the use of state chart diagram? Show mutual exclusive condition in Collaboration diagram with example. 7
5.

a) How DCD is different from Domain models? Explain method name issues in DCD. 7

b) Define GRASP. Explain the purpose of Controller pattern with an example? 8
6. CASE STUDY: Draw equivalent use case, sequence diagram and class diagram for the given scenario below. 15

A customer wants to draw money from his bank account. He enters his card into an ATM (automated teller machine). The ATM machine prompts "Enter PIN". The customer enters his PIN. The ATM (internally) retrieves the bank account number from the card. The ATM encrypts the PIN and

the account number and sends it over to the bank. The bank verifies the encrypted Account and PIN number. If the PIN number is correct, the ATM displays "Enter amount", draws money from the bank account and pays out the amount.

7. Write short notes on: (Any two)

- a) Object diagram
- b) Creator pattern
- c) UML multiple perspectives

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Attempt all the questions.

1. (a) What is UML? How agile is applied to software development process? 7
- (b) What is object oriented analysis and design? Describe the most used software development life cycle used nowadays? 8
2. Case Study: Bank Transaction System:

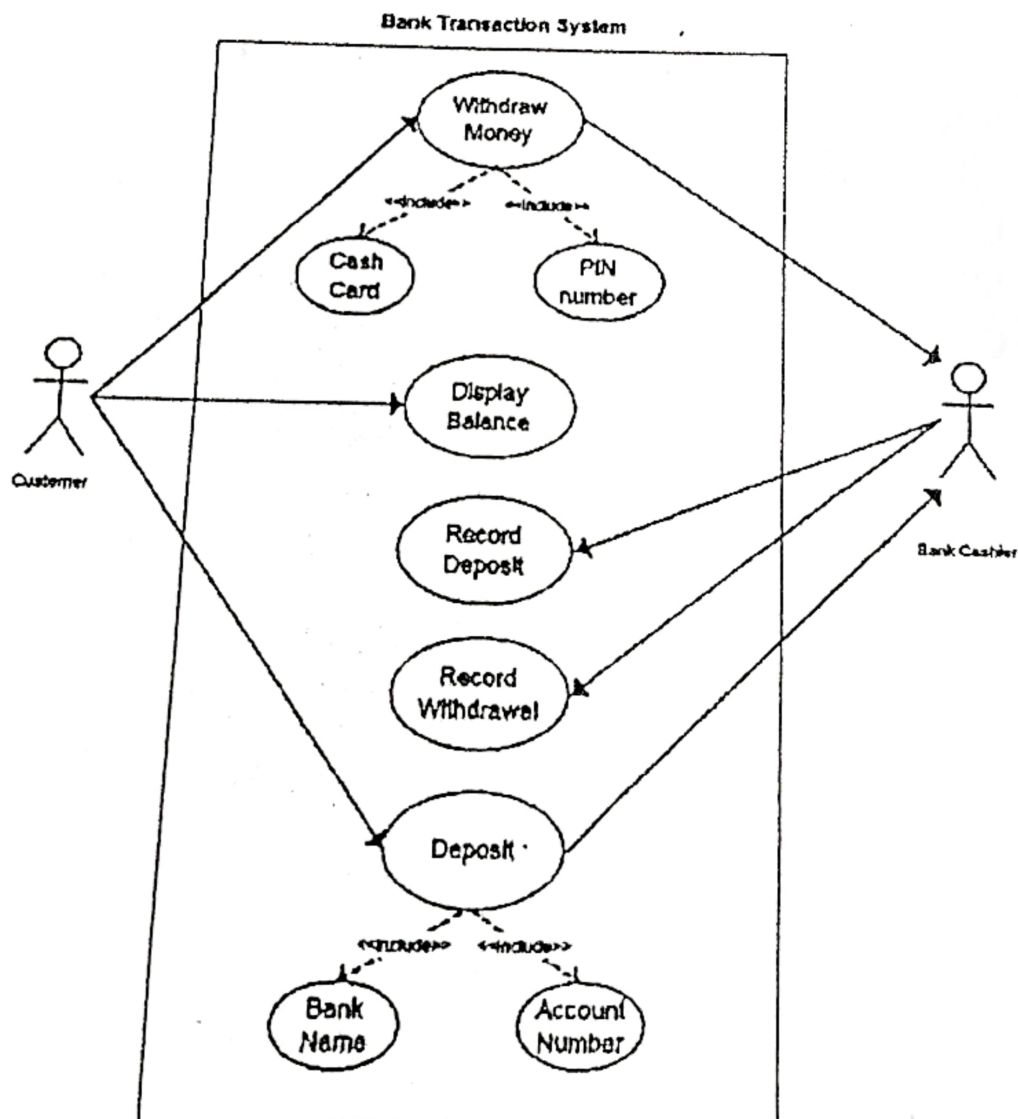


Fig: use case of Bank Transaction System

- On the basis of above Use case:
- a) Develop the system sequence diagram for Withdraw Money and Deposit functions. 7
 - b) Write the contract for any one of the system operations you identify. 8
 3. a) What is conceptual Model used for? How system behaviour can achieve data abstraction and Encapsulation? Explain. 7
 - b) When contracts are used? Explain in context of an example by using the standard guidelines. 8
 4. a) What does visibility controls in Object oriented programming? Describe different types of it with example. 7
 - b) What are the roles of objects and patterns in OODM? Describe the different patterns used. 8
 5. a) How domain model can be built from use case? Illustrate the steps used. 8
 - b) What are the important things to be handled during mapping design to code? Illustrate with an example. 7
 6. a) Why exception and error handling is important in OOAD? Explain. 7
 - b) What are descriptor classes in GRASP? How does it achieve singleton functionality? 8
 7. Write short notes on: (Any two) 2×5
 - a) Use Case Model
 - b) MVC Pattern
 - c) Creating methods from collaboration diagram

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Time : 3hrs.

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Attempt all the questions.

1. You are developing a system to handle checkout duties at a book and video store. Sales boy may checkout books and videos from the store. The system must keep track of which books and videos have been sold out.

Each category of book and video has a unique item number that is used by the system to index that specific book or video, a sales boy may check the status of a book (in stock or not) by supplying the item number. He may also check either on a book by supplying the title and author or on a video by supplying only the title.

For the above case study:

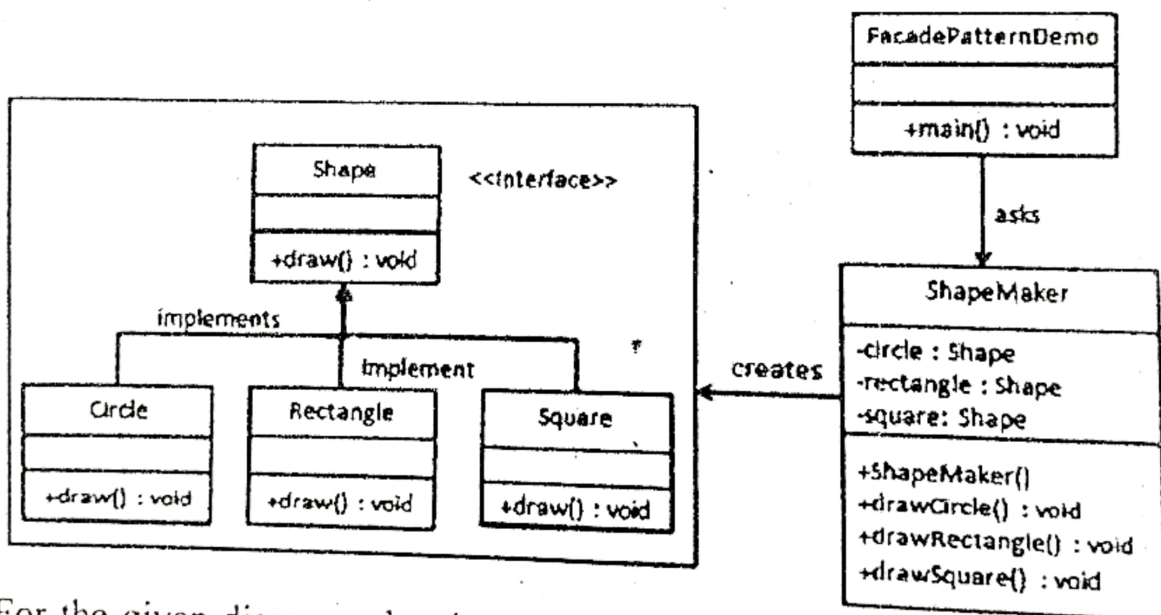
- a) List use cases and prepare use case diagram. 8
- b) Write down any one essential use case text from the above use case diagram. 7

2. Case Study 15

Consider an online travel agency system. A new user can create a new profile with a username and a password. This profile can be changed at any time if the user wants. The user logs to the system using his/her username and password. After logging into the system, the user inquires about different flights. The user can book, change or cancel a flight reservation. This includes choosing a payment method, and making a payment.

- a) Now draw a use case model (no description required), Domain model,

3. a) interaction diagrams and design class diagram from these information. 7
 How are system event and system operation important. Assume the scenario of reserving a hotel room online and draw a SSD for it.
- b) Why are contracts essential? Assume the scenario of reserving a hotel room online and draw a contract for it. 8
4. a) What are interaction diagrams? What are its types? How is sequence diagram similar to collaboration diagram? 8
 b) What are the GRASP patterns? Explain with an example how you apply the creator pattern during object design. 7
5. a) What is visibility? Describe the use of visibility among classes while designing class diagram. 8
 b) How procedure Analysis and Design is different from OOAD, illustrate with examples? 7
6. a) Describe the importance of interaction diagrams for developing methods during coding. 5
 b) 10



For the given diagram, develop the equivalent code in any high level programming language.

7. Write short notes on: (Any two)
 - a) OO Development Cycle
 - b) Associations and Attributes
 - c) Patterns

2×5

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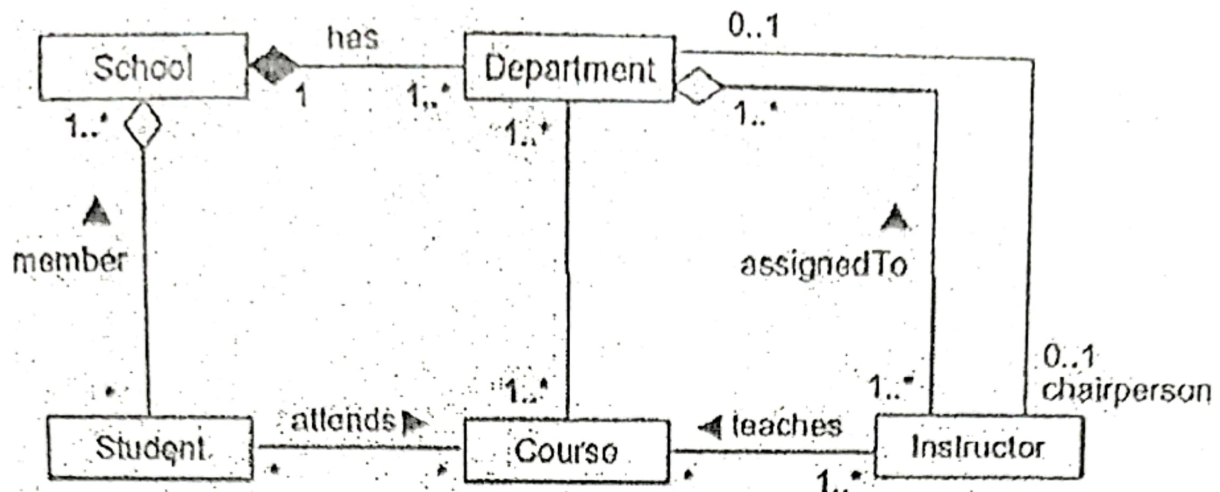
Time : 3hrs.

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Attempt all the questions.

1.
 - a) What is the role of notations in a modelling tool? Illustrate any five different notations used in UML with brief notes of each. 3+5
 - b) In OOAD, there are various types of models, like conceptual, structural, behavioural, etc. What is the significance of these many different types of model? Explain with illustrative examples. 3+4
2. Imagine you are hired as IT consultant for the task of building Pokhara University Exam Management System (PU-EMS), considering your four-year/8-semester exposure of the overall university board exam as well as college exams. As a domain study, you have found out there is the provision of double checking of all the questions and copies to ensure the integrity and standard of all of the processes. 8+7
 - a) Prepare a brief SRS document of PU-EMS with all details as per SRS document.
 - b) Draw the Use-case diagrams for this system with proper stick diagrams.
3.
 - a) What is the role of constraint specifications in defining the systems behavior model diagram? The model below depicts about possible types of accounts within the portfolio sub-head in any banking business domain. Identify the constraints highlighted and explain in brief, what they mean. 2+5



b) Prepare the class diagram for the SRS and Use case diagrams that you have already developed for Question No. 2 system.

7

6. Compare the followings:

3×5

- X-axis (Left-right) versus Y-axis (top-down) Information ordering in Sequence (Interaction) diagram
- Forking versus Joining in Activity diagram
- Structural versus behavioral patterns in design reuse

7. Write short notes on: (Any two)

2×5

- Reverse Engineering in OOD
- Analysis using CRC
- Exception handling through UML